PROCESS PARAMETERS AND DIAGNOSTIC GUIDE

AN OPERATION SUPPLEMENT MANUAL FOR THE

CINCINNATI CL-707 AND CL-7A C4000 LASER SYSTEMS

CINCINNATI

CINCINNATI INCORPORATED
CINCINNATI, OHIO 45211

EM-512 (N-04/05)

COPYRIGHT © 2005
CINCINNATI INCORPORATED

PROCESS PARAMETERS GUIDE CONTENTS

INTRODUCTION

SECTION A PROCESS NOTES A-1

GENERAL COMMENTS A-1
CUTTING PARAMETERS AND CUTTING TECHNIQUES A-1
MISCELLANEOUS COMMENTS A-1
MILD STEEL A-4
N_2 ASSIST A-4
O_2 ASSIST, DRY A-5
STAINLESS STEEL (TYPE 304) A-7
N_2 ASSIST A-7
ALUMINUM (5052-H32) A-9
N_2 ASSIST A-9

SECTION B PARAMETER TABLES B-1

MILD STEEL B-1
N_2 ASSIST B-1
O_2 ASSIST, DRY B-2
STAINLESS STEEL (TYPE 304) B-4
N_2 ASSIST B-4
ALUMINUM (5052-H32) B-6
N_2 ASSIST B-6

SECTION C PROGRAMMING AID C-1

MILD STEEL C-1
N_2 ASSIST C-1
O_2 ASSIST C-2
STAINLESS STEEL (TYPE 304) C-4
N_2 ASSIST C-4
ALUMINUM (5052-H32) C-6
N_2 ASSIST C-6

SECTION D DIAGNOSTIC GUIDE D-1

GENERAL DIAGNOSTIC GUIDELINES D-1
ALUMINUM D-2
MILD STEEL AND TOOL STEEL – O_2 ASSIST D-3
MILD STEEL, GALVANIZED, AND STAINLESS STEEL – N_2 ASSIST D-6
COPPER ALLOYS D-7
COMPOSITES AND PLASTICS D-8
WOOD D-9
LASER CUTTING DIAGNOSTIC WORKSHEET D-10

SECTION E MACHINE EVALUATION E-1

CHECKING POWER STABILITY E-1
CHECKING BEAM ALIGNMENT E-1
CHECKING MODE QUALITY E-2
TABLE MODE BURN EXAMPLES E-3

COPYRIGHT © 2005
CINCINNATI INCORPORATED

INTRODUCTION

PROCESS PARAMETERS AND DIAGNOSTIC GUIDE

The purpose of this manual is to help users of CINCINNATI INCORPORATED laser machining systems prevent many common laser cutting problems and quickly fix those problems (in the cutting process) that do occur.

This manual consists of five sections:

- SECTION A – PROCESS NOTES; includes comments, hints, and suggestions regarding the cutting parameters. The first process notes page contains general process information. Each of the remaining process notes pages contains information about cutting a particular material and matches one of the parameter tables in SECTION B.
- SECTION B – PARAMETER TABLES; provides the cutting parameters (parameter tables) for most common cutting applications.
- SECTION C – PROGRAMMING AID; provides useful information regarding attributes used in the Laser Nesting Software to assist with the programming of materials listed in this guide.
- SECTION D – DIAGNOSTIC GUIDE; provides troubleshooting assistance. General diagnostic guidelines are provided as well as detailed diagnostic charts for each of several material categories. A laser diagnostic worksheet is also included in this section.
- SECTION E – MACHINE EVALUATION; provides information that will assist trained maintenance personnel. Information is provided for checking power stability, beam alignment, and mode quality.

If a particular laser-cutting problem cannot be solved using the procedures in this manual, document the problem using the LASER CUTTING DIAGNOSTIC WORKSHEET in SECTION D. Refer to this worksheet when calling the CINCINNATI INCORPORATED Laser Applications Department for assistance. Note, the phone number is on the worksheet.

Note: *Detailed diagnosis of machine problems and methods for fixing problems are not included in this manual. Contact CINCINNATI INCORPORATED, Laser Service Department for proper maintenance of your laser machining system.*

GENERAL COMMENTS

CUTTING PARAMETERS AND CUTTING TECHNIQUES

The laser cutting parameters in Section B are guidelines. They have been developed to help you, the laser user, obtain consistent cutting performance in daily operation with a well-adjusted Cincinnati Incorporated Laser System. These parameters were established using commercially available 1008/1010/1020 mild steel (cold rolled, HRPO, or laser-quality hot-rolled), type 304-2B stainless steel, and type 5052-H32 mill-finished aluminum. All materials had clean, smooth and consistent surface condition.

You may choose to adjust some of these parameters to optimize cutting performance for your application. Several factors could lead you to make parameter adjustments. If your material composition and/or surface conditions are different from those used in developing these parameters, some adjustments are likely. In this case, use the parameters for a similar material of the same thickness as a starting point. If a particular part has very closely spaced features or intricate contours, some minor adjustments might be helpful. Usually these adjustments would be designed to control localized heating of the workpiece. Finally, you might wish to increase feedrates above the recommended values. Sometimes this feedrate increase is achievable by "tweaking" the cutting parameters and (possibly) sacrificing consistency of the edge quality.

The following notes provide additional information regarding these parameters, and they describe helpful cutting techniques.

MISCELLANEOUS COMMENTS

1. Focus setting is the vertical position of the focal point relative to the nozzle tip (not the workpiece). For the manual focus cutting head, set the focus by reading the scale on the head (after the focus scale has been calibrated). The focus setting for the Auto Focus head is in the cutting parameter library. For calibrating the zero focus reference position of either cutting head refer to the *CL-707* or *CL-7A Operation, Safety, and Maintenance Manual* for details.

2. Machines with an Auto Focus cutting head require less manual intervention. In addition, lowering the pierce focus setting may reduce pierce times. The focus setting in the cutting parameter tables are based on a manual focus head and reference the far field position of the table. Since the middle of the focus window in the near field is lower (more negative) than the middle of the far field focus window, a lower cut focus setting may be used in the near field and a slightly higher focus setting may be used in the far field to help optimize the cutting performance.

3. Process parameters were developed using the recommended nozzle tips listed below. The 125sp nozzle has an exit orifice of 0.125" (3.2 mm) and is 0.100" (2.5 mm) shorter than the other nozzle tips. This nozzle is used for nitrogen assist cutting of thick stainless steel and aluminum. It is shorter to achieve a more negative focus than the cutting head will allow with the standard length tips.

   The design of the High Capacitance (hc) nozzle improves the consistency of the cut standoff especially when plasma is present. Consistency of the cut standoff is vital when processing thin materials at high speed with nitrogen or air assist. Deviating from these nozzle tips may require parameter adjustments to achieve consistent results.

   ◊ 0.040" CI #924635

   ◊ 060hc (High Capacitance) CI #924636

   ◊ 080hc (High Capacitance) CI #924637

   ◊ 040shr (Shower nozzle)* CI #923013

   ◊ 062shr (Shower nozzle) CI #923014

   ◊ 100str (Long, straight section) CI #924638

   ◊ 125sp (Special, 0.100" shorter) CI #925127

   **As an alternative to the 062shr nozzle, the 040shr nozzle can be used for processing 0.375" (9.5 mm) and 0.5" (12.7 mm) laser quality hot rolled steel.*

4. When ramped pierce is recommended, the pierce time shown in the table is equal to the sum of the step times plus the cooling time. Piercing in the near field generally takes longer than piercing in the far field of the optical path length. This difference is more apparent when piercing thick mild steel. When using the Auto Focus head, the pierce times can be made similar at the near field (NF) and far field (FF) locations by lowering the NF pierce focus compared

to the FF pierce focus. Lowering the pierce focus position (when using the Auto Focus cutting head) may reduce ramped pierce times.

5. Use Extended Pierce Cycle whenever the 0.040" (non-shower) nozzle is used. Extended Pierce Cycle is used to optimize the initial start of cut during the first pierce. The extended pierce cycle is set in the configuration page under [Maintenance] – [Configuration]. The settings are: Beam Off time=10 seconds, Extra Cycle Time=1.0 second. In some cases an Extra Cycle time of up to 2.0 seconds may be beneficial. Turn this feature off when cutting plastic coated stainless steel (set Beam Off time=0). The "Supervisor" password is required to edit the settings for this feature.

6. The 0.040" (non-shower) nozzle is recommended for cutting HRPO mild steel with oxygen assist from 0.188" (4.8 mm) to 0.375" (9.5 mm). For this cutting process to work well use these techniques together: extended pierce cycle, head down operation, speed gas on, and rapid pierce. Dross that comes off just by touching it is called 'crumble dross'. This kind of dross is typical when using the 0.040" nozzle.

7. For mild steel oxygen assist, the 0.040" nozzle is beneficial in the following ways; it reduces pierce time, it produces excellent edge quality, it allows higher feedrates, and it uses less assist gas.

8. Optional pressure (M67) enables the use of a second cutting assist gas pressure. For example, you can use the cut pressure to start the cut and use the optional pressure for the rest of the cutout. The optional pressure feature is used primarily for nitrogen assist cutting of stainless steel, mild steel, and aluminum. A dwell time of 300 to 500 msec is necessary after the M67 to achieve a smooth cut edge immediately after the lead-in. Optional pressure may also be used for oxygen assist cutting of thick mild steel in conjunction with rapid pierce. This pressure can be different in the near field and far field areas to be used like a dynamic-optional pressure.

9. Nitrogen assist gas is preferred over oxygen for cutting thin material. For example, higher feedrates can be achieved for 11 gage (3.0 mm) mild steel and thinner. In addition to higher cutting speeds, the cut edge is free of oxidation unlike the cut edge when using oxygen or air assist.

10. Execution of the G89 command will set the feedrate variable #148 (if configured to do so). Actual cut feedrate is limited by accuracy requirements for small features and is calculated within the nesting (CAM) software.

11. The "Z hold distance" parameter enables a function that improves the cutting process for nitrogen or air assist. After completing a pierce, the Z-axis moves the nozzle to the Cut standoff position. When "Z hold distance" is not zero, the Z-axis position does not change again until the X and Y-axes have moved the specified distance away from the pierce location. Without this function, the Z-axis would change position in response to the plasma initially generated by nitrogen or air assist cutting, producing incorrect nozzle standoff.

    "Z hold distance" is not used when cutting with oxygen assist because the height of the pierce mound makes the initial Cut standoff too high. If the Z-axis maintained that high standoff, the cutting process might not start correctly and could degrade to thermal runaway. When cutting with oxygen assist, set "Z hold distance" to zero.

    If a program commands Optional Standoff (M45) when the X Y position is inside the "Z hold distance", the Z-axis will change to normal tracking mode, maintaining the Optional Standoff setting.

12. Minimum processing time is obtained by cutting with the head down between features and turning the Speed Gas option ON. Also, a zero pierce time will save 0.080 to 0.100 seconds per pierce.

13. Process parameters were developed with a focus window overlap of 0.025" (0.6 mm) from near field to far field.

14. Optional standoff (M45) is used when cutting medium to thick mild steel, stainless steel, or aluminum with nitrogen assist gas. This function allows the lead-in to be cut at a higher standoff than the feature being cut. When optional standoff is recommended, the lead-in standoff (specified with the Cut standoff parameter) is typically 0.100" (2.5 mm) and the optional standoff is typically 0.025" (0.6 mm).

    The program cuts the lead-in using the (higher) "Cut" standoff to protect the nozzle tip, and then commands M45 so the rest of the cutting path uses the (lower) Optional standoff. However, at the lower standoff distance, the nozzle tip can be damaged if the height sensor moves the Z-axis in response to debris on the material surface. To avoid nozzle damage, the programmer should select a pierce location far away from the final cutting path, so the nozzle can complete the cutting path without the Z-axis responding to debris left by the pierce. Optional Pressure (M67) is generally used together with Optional Standoff (M45).

Note that "optional standoff" and "optional pressure" are standard features of the CL-707 and CL-7A. The work "optional" in these cases refers to a second, or alternate, value.

MILD STEEL

N_2 ASSIST

The use of nitrogen assist prevents the creation of an oxide layer on the cut edge. Recommended parameters are given for thickness up to 0.25" (6.4 mm). Nitrogen assist is not recommended for thickness over 0.25" (6.4 mm) because the feedrate is considerably lower than with oxygen assist.

1. Pierce time – Use a zero pierce time for 18 gage (0.048", 1.2 mm) and thinner. This setting minimizes the part processing time but may cause undesirable cosmetic imperfections on small features. By using a non-zero pierce time, such as 0.010 seconds, these imperfections can be minimized.

2. Pierce power – The recommended piercing power is 3000W. Piercing with a power setting of more than 3000W may cause lens spatter.

3. Pierce duty cycle – A pierce duty cycle of 20% is recommended to produce a small pierce hole and minimize the size of the pierce mound. The benefit of smaller pierce holes is that the lead-in length can be minimized.

4. Pierce standoff – As material thickness increases, an increasingly higher pierce standoff is recommended. The purpose of the higher standoff is to get a small spot size for the pierce and to reduce possible lens damage. The higher standoff also allows clearance for the pierce mound created during the pierce process with thick material.

5. Pierce pressure – If the pressure is too low, lens damage is more likely. If the pressure is too high, plasma formation is likely which will increase the pierce time.

6. Ramped pierce – Ramped pierce is not needed when cutting with nitrogen assist gas.

7. Cut feedrate – Use the recommended feedrate for most features. When cutting small features, a reduced feedrate might be necessary to maintain contouring accuracy.

8. Pre-cut dwell – Pre-cut dwell is not needed for processing mild steel with nitrogen assist.

9. Cut power – Use the maximum power at 100% duty cycle for cutting with nitrogen assist gas to get maximum feedrate.

10. Cut frequency – Use 2000 Hz for best performance.

11. DPC feedrate – Use 100% of the maximum cutting feedrate for 22 gage (0.030", 0.8 mm) to 10 gage (0.135", 3.4 mm) material. This helps reduce the dross on sharp corners. If a loss of cut occurs while exiting sharp corners, set the DPC feedrate below the maximum cutting speed.

12. Cut standoff – A low cut standoff is recommended to get the best assist gas flow through the kerf. See comments on optional cut standoff below.

13. Nozzle – Use the recommended 060hc, 080hc, or 100str nozzle. Do not use a shower nozzle for nitrogen cutting because it does not improve cutting performance, and will increase assist gas consumption.

14. Optional standoff – Optional standoff is recommended for 13 gage (0.090", 2.3 mm) to 0.25" (6.4 mm) material to prevent the pierce mound from damaging the nozzle tip. Use M45 at the end of the lead-in to activate the optional standoff. When optional standoff is recommended, the cut standoff is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M45 command after the lead-in.

15. Optional assist gas pressure – Optional pressure is recommended for all mild steel cutting with Nitrogen assist. Use M67 at the end of the lead-in to activate the optional assist gas pressure. This process uses the same pressure for near field and far field cutting. When optional pressure is recommended, the assist gas pressure is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M67 command after the lead-in.

16. Start-of-cut process – To initiate the cut, use low cut pressure and low feedrate (10% to 30% of cut feedrate) during the lead-in. Increase the lead-in length 0.050” (1.3 mm) to 0.200” (5.1 mm) as thickness increases. Use M45 to activate optional standoff together with M67 to activate optional pressure. Use a 0.200 to 0.500 second dwell after the M67 to help transition to the cut process. Continue at the recommended feedrate.

MISCELLANEOUS COMMENTS

1. Large hard dross can be eliminated by lowering focus, increasing nitrogen assist gas pressure, and/or increasing cutting speed.

2. Fine (sandpaper type) dross can be eliminated by lowering the cutting speed, lowering the nitrogen assist gas pressure and/or raising the focus.

3. Add or increase the dwell time after the M67 (optional assist gas) to eliminate a rough cut edge after the lead-in.

O_2 ASSIST, DRY

For mild steel up to 0.50” (12.7 mm) thick, the listed parameters apply to CR (cold-rolled) and HRPO (hot-rolled, pickled and oiled) steel. Since HRPO treatment is expensive for thicker mild steel, the listed parameters for thickness above 0.50” (12.7 mm) apply to laser-quality hot-rolled (HR) steel.

When cutting laser quality HR (0.50" (12.7 mm) and thinner) instead of HRPO, some parameter changes might be necessary to prevent thermal runaway. Some changes to try are: reduce assist gas pressure, use the 040shr nozzle instead of the 062shr nozzle, reduce power, and reduce feedrate.

1. Pierce power – Use the recommended piercing power. Exceeding the recommended piercing power may cause blowouts and/or lens spatter.

2. Pierce standoff – Use a 0.080" (2.0 mm) standoff or greater for all materials.

3. Pierce time – When using an auto focus head, a lower pierce focus can be used to reduce the pierce time.

   The parameter table indicates pierce times in several rows. When Ramped Pierce is specified, the “normal” pierce time (G89 D) indicates the total time for the ramped pierce steps. When the application includes Rapid Pierce parameters, “normal” pierce time is listed with a second value indicating the sum of rapid pierce Beam On and Cool times.

4. Pierce pressure – Recommended pressure is relatively low to avoid blowouts. If the pressure is too low, lens damage is more likely.

5. Ramped pierce – Use ramped pierce for 10 gage (0.135", 3.4 mm) and thicker to prevent blowouts. If blowouts occur, the recommended action depends on the type of blowout. There are two common types of pierce blowouts. The first is a blowout before the pierce time has elapsed. This type can be corrected by lowering the pierce duty cycle, power, or assist gas pressure. As a last resort, lengthen the ramp time.

   The second type of blowout occurs after the pierce time has elapsed, and is caused by an incomplete pierce within the specified pierce time. A blowout of this nature can be corrected by increasing the pierce duty cycle, power, or assist gas pressure. If these changes cause the first type of blowout, lengthen the hold time instead.

6. Cooling time – Cooling time is recommended for thick mild steel (0.5" (12.7 mm) and thicker). This parameter helps control the temperature for the start-of-cut, and may reduce gouging at the start-of-cut for small features.

7. Cut feedrate – For large features, use the recommended feedrate. For small features (close to material thickness in size), use 80% of recommended feedrate.

8. Pre-cut dwell – This parameter turns the beam on for the specified amount of time before the cut process begins. When the power burst time is zero, the power level during the pre-cut dwell corresponds to the DPC min value. When the power burst time is greater than zero, the power level during the pre-cut dwell corresponds to the full cut power.

   For thick mild steel 0.188" (4.8 mm) and thicker, where the pierce standoff is higher than the cut standoff, a short pre-cut dwell may help provide a cleaner start of cut. A pre-cut dwell is also used to allow time for all other machine cut parameters to stabilize.

9. Cut power – Recommended power is lower than maximum available power for 10 gage (0.135", 3.4 mm) and thinner. If the power is too high, overheating of the material will cause poor edge quality, especially for small features and at corners.

10. DPC feedrate – Set to 70% to 85% of cut feedrate when using DPC. This setting improves cut quality for small features cut at a reduced speed. It also helps prevent a tab at the end of a feature when ramping down the feedrate. Some materials do not use DPC.

11. Cut standoff –Use a 0.040" (1.0 mm) standoff when using shower nozzles. Use a 0.050" (1.3 mm) standoff for the 060hc and 080hc nozzles. Use a cut standoff of 0.030" (0.8 mm) for the 0.040" nozzle.

12. Kerf width – The kerf width is a good indicator of the proper focus setting when cutting mild steel. If the measured kerf is less than the value shown in the parameter table, then raise the focus.

13. Nozzle – Use the recommended 0.040", 060hc, or 062shr nozzle for processing mild steel with oxygen. The 040shr nozzle can be used for cutting up to 0.5" (12.7 mm) thick material.

14. Lead-in – Use a short lead-in, 0.020" (0.5 mm) to 0.050" (1.3 mm), for 0.25" (6.4 mm) and thicker with a ramped pierce. Cut the lead-in at 70 to 80% of the recommended cut feedrate. Short lead-ins reduce overheating of small features, and contribute to a better start of cut. For thinner materials, lead-in length is less critical, and can be anywhere from 0.030" (0.8 mm) to 0.080" (2.0 mm).

15. For optimum results, the minimum dimension of small features must be greater than or equal to the material thickness. To cut smaller features in thick mild steel, contact the Laser Applications department for "small feature" cutting techniques and parameters.

16. Rapid Pierce – Assist gas pressure should be low (8 to 10 psi) to reduce the velocity of spatter particles during the pierce. Use a high standoff of 0.200" (5.1 mm). The high standoff prevents lens damage and produces a larger pierce hole, allowing complete removal of the pierce mound.

    The air-blast off time should be approximately equal to the actual pierce through time. If the off time is too short, the air-blast will distort the pierce hole. If the off time is too long, the pierce mound will cool and be more difficult to remove.

    A longer lead-in (at least 1/2 material thickness) is needed when rapid pierce is used.

    If rapid pierce is recommended, a second pierce time is shown in the parameter table. The time shown is the sum of the beam on time and the cool time.

STAINLESS STEEL (TYPE 304)

N_2 ASSIST

Stainless steel is commonly cut with nitrogen assist gas. The use of nitrogen prevents the creation of an oxide layer on the cut edge. It also produces a considerably smoother and more consistent edge quality. Recommended parameters are given up to 0.5" (12.7 mm).

1. Pierce time – Use a zero pierce time for 18 gage (0.048", 1.2 mm) and thinner. This setting minimizes the part processing time.

2. Pierce power – For 11 gage (0.120", 3.0 mm) down to 16 gage (0.060", 1.5 mm), 3000W is the recommended pierce power. For 18 gage (0.048", 1.2 mm) and thinner use 2700W. Increasing the power for these materials may cause lens spatter. For 10 gage (0.135", 3.4 mm) and thicker, use 4000W to keep pierce times short.

3. Pierce duty cycle – For 11 gage (0.120", 3.0 mm) and thinner, the recommended pierce duty cycle is 20%. Duty cycles above 30% will likely cause lens spatter.

4. Pierce standoff – As material thickness increases, an increasingly higher pierce standoff is recommended. The purpose of the higher standoff is to get a small spot size for the pierce and to reduce possible lens damage. The higher standoff also allows clearance for the pierce mound created during the pierce process with thick material.

   Pierce standoff is ignored if the pierce time is zero.

5. Pierce pressure – If the pressure is too low, lens damage is more likely. If the pressure is too high, plasma formation is likely which will increase the pierce time.

6. Ramped pierce – Ramped pierce is not used when cutting with nitrogen assist gas.

7. Cut feedrate – Use the recommended feedrate for most features. When cutting small features, a reduced feedrate might be necessary to maintain contouring accuracy.

8. Cut power – Use maximum power for cutting with nitrogen assist gas to get maximum feedrate.

9. Cut frequency – A cut frequency of 2000 Hz is recommended for all materials. When cutting at full feedrate, the cut frequency makes little difference since the laser is set to 100% duty cycle. However, the frequency has more of an affect on the cut when the feedrate is reduced at corners and small features. The DPC feedrate and min power % needed to obtain good edge quality depend on the cut frequency. The DPC parameters were optimized for a cut frequency of 2000 Hz.

10. Cut standoff – A low cut standoff is recommended for all materials to get the best assist gas flow through the kerf. See the following comment on optional standoff.

11. Nozzle – Use the 060hc, 080hc, 100str, or 125sp nozzles as recommended. Testing of shower nozzles for nitrogen cutting has shown no improvement in the cutting performance, but has shown increased assist gas consumption.

12. Optional assist gas pressure – Optional pressure is recommended for 18 gage (0.048", 1.2 mm) and thicker. Use M67 at the end of the lead-in to activate the optional assist gas pressure followed by a dwell time to allow the assist gas pressure to stabilize. When optional pressure is recommended, the assist gas pressure is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M67 command after the lead-in.

13. Optional standoff – Optional standoff is recommended for 13 gage (0.090", 2.3 mm) and thicker to prevent the pierce mound from damaging the nozzle tip. Use M45 at the end of the lead-in to activate the optional standoff. When optional standoff is recommended, the cut standoff is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M45 command after the lead-in.

14. Start-of-cut process – Use a slow feedrate (10% to 30% of cut feedrate) during the lead-in for all nitrogen cut stainless steel applications. Increase the lead-in length 0.050" (1.3 mm) to 0.350" (8.9 mm) as thickness increases. Use optional standoff (M45) together with optional assist gas (M67) at the end of the lead-in. Use a 0.300 to 0.500 second dwell and/or ramp up the feedrate in several segments after the M67 to help transition into the cut process. Continue at the recommended feedrate.

    A combination of a 0.300 second dwell and a ramped start of cut after the M67 produce a very stable process in thicker materials.

    Using a ‘ramped start-of-cut’ can minimize the plasma formation at the start of the cutting process (after the lead-in and M67). Using the minimum

required assist gas pressure will also minimize plasma formation.

Miscellaneous comments

1. Coatings – For thin stainless steel that is plastic coated, a slight feedrate reduction may be necessary to optimize edge quality. For some PVC coated stainless steel, the coating must first be cut to prevent "bubbling" of the coating. For more techniques and information on cutting PVC coated stainless contact the Laser Applications department.

2. Consistent long "icicle" type dross can be eliminated by lowering focus, increasing nitrogen assist gas pressure, and/or increasing cutting speed.

3. Fine (sand paper type) dross can be eliminated by lowering the cutting speed, lowering the nitrogen assist gas pressure, and/or raising the focus.

4. Increase the dwell time after an M67 to eliminate a rough cut edge after the lead-in.

ALUMINUM (5052-H32)

N_2 ASSIST

The cut edge is dross-free up to 0.080" (2.0 mm). For thicker materials, a light dross is formed that should break off easily.

1. Pierce time – Use a zero pierce time for 0.063" (1.6 mm) and thinner.

2. Pierce power – A pierce power setting of 4000W is recommended whenever cutting aluminum.

3. Pierce frequency – Programs for applications with frequency below 100 Hz must specify parameters with a library file. The explicit G89 macro cannot specify frequency below 100 Hz.

4. Pierce and cut assist gas – Nitrogen is used on all materials.

5. Pierce duty cycle – The pierce duty cycle is set to 100% to provide the fastest piercing times in thin materials. In thicker materials, high duty cycle causes large pierce mounds. For 0.188" (4.8 mm) and thicker, reduced frequency and duty cycle are used to control the pierce mound size.

6. Pierce standoff – As material thickness increases, an increasingly higher pierce standoff is recommended. The purpose of the higher standoff is to get a small spot size for the pierce and to reduce possible lens damage. The higher standoff also allows clearance for the pierce mound created during the pierce process with thick material.

Pierce standoff is ignored if the pierce time is zero.

7. Pierce pressure – If the pressure is too low, lens damage is more likely. If the pressure is too high, plasma formation is likely which will increase the pierce time.

8. Ramped pierce – Ramped piercing is not used for aluminum.

9. Cut feedrate – Use the recommended feedrate for most features. When cutting small features, a reduced feedrate might be necessary to maintain contouring accuracy.

10. Pre-cut dwell – This parameter turns the beam on at DPC min for the specified amount of time. This function is used only on 0.5" (12.7 mm).

11. Cut power – Use maximum power for cutting to get maximum feedrate.

12. Dynamic power control – Dynamic power control is not used for thick aluminum (0.313" (8.0 mm) or greater) since loss of cut at the corners and start of cut problems can occur.

13. DPC feedrate – Setting this feedrate to 90% of the maximum cutting feedrate gives better power control through areas of acceleration and deceleration.

14. Cut standoff –Use a cut standoff of 0.025" (0.6 mm) for all materials to get the best assist gas flow through the kerf. Also, see comments on optional standoff below.

15. Nozzle – Use the 060hc, 080hc, 100str, or 125sp nozzles as recommended. A shower nozzle is not recommended for cutting aluminum because it does not improve cutting performance, but does increase assist gas consumption.

16. Optional assist gas pressure – Optional pressure is recommended for 0.080" (2.0 mm) and thicker. Use M67 at the end of the lead-in to activate the optional assist gas pressure. This process uses the same pressure for near field and far field cutting. When optional pressure is recommended, the assist gas pressure is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M67 command after the lead-in.

17. Optional standoff – Optional standoff is recommended for 0.090" (2.3 mm)and thicker to prevent the pierce mound from damaging the nozzle tip. Use M45 at the end of the lead-in to activate the optional standoff. When optional standoff is recommended, the cut standoff is shown with an asterisk (*) in the parameter table as a reminder that the program should include an M45 command after the lead-in.

18. Start-of-cut process – Use a slow feedrate (10% to 30% of full cut feedrate) during lead-in for all aluminum applications. Increase the lead-in length 0.050" (1.3 mm) to 0.400" (10.0 mm) as thickness increases. Use M45 (optional standoff) at the end of the lead-in for materials 0.090" (2.3 mm) and thicker. Use M67 (optional pressure) at the end of the lead-in for materials 0.080" (2.0 mm) and thicker. Use a 0.300 to 0.500 second dwell and/or ramp up the feedrate in several segments after the M67 to help transition into the cut process. Continue at the recommended feedrate.

MILD STEEL

N_2 ASSIST

	Thickness (inch)			**0.030**	**0.036**	**0.048**	**0.060**	**0.075**	**0.090**	**0.105**	**0.120**	**0.135**	**0.188**	**0.250**	
	Thickness (gage)			22	20	18	16	14	13	12	11	10	3/16	1/4	
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q		3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	3000	
	Frequency (Hz)	G89 R		200	200	200	200	200	200	200	200	200	200	200	
	Duty cycle (%)	G89 R		20	20	20	20	20	20	20	20	20	20	20	
	Standoff (inch)	G102 S		0.050	0.050	0.050	0.050	0.050	0.080	0.080	0.100	0.100	0.100	0.100	
	Pierce time (sec)	G89 D		**0.00**	**0.00**	**0.00**	**0.01**	**0.01**	**0.05**	**0.05**	**0.10**	**0.20**	**0.50**	**0.70**	
	Z hold distance (inch)			0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	
	Part coolant (on/off)	G89 J		9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	
	Ramped pierce (on/off)	G103		off	off	off	off	off	off	off	off	off	Off	off	
	Assist gas port	G89 B		12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	
	Assist gas press (psi)	G89 E		30	30	30	30	30	30	30	30	30	30	30	
Rapid Pierce	Laser power (Watt)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Standoff (inch)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Pierce time (sec)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Cool time (sec)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Airblast off time (sec)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Airblast on time (sec)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Assist gas press (psi)	n/a		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
Ramped Pierce	Number of steps	G103 Q		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 0 power (% max)	G103 R		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 1 time (sec)	G103 A		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 1 power (% max)	G103 S		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 2 time (sec)	G103 B		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 2 power (% max)	G103 T		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 3 time (sec)	G103 C		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Step 3 power (% max)	G103 U		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Cooling time (sec)	G103 F		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
Cutting	Laser power (Watt)	G89 T		4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	
	Frequency (Hz)	G89 S		2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	
	Duty cycle (%)	G89 S		100	100	100	100	100	100	100	100	100	100	100	
	Pre-cut dwell (sec)	G102 D		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	Feedrate (ipm)	F #148		**450**	**420**	**290**	**320**	**290**	**260**	**200**	**160**	**135**	**90**	**62**	
	Kerf width (inch)	G89 K		0.008	0.008	0.008	0.009	0.009	0.009	0.009	0.009	0.010	0.011	0.012	
	Standoff (inch) [NC]	G102 Z		0.025	0.025	0.025	0.025	0.025	0.100*	0.100*	0.100*	0.100*	0.100*	0.100*	
	Optional standoff (inch)	M45		n/a	n/a	n/a	n/a	n/a	0.025	0.025	0.025	0.025	0.025	0.025	
	Coolant (on/off)	G89 C		9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	
	Dynamic pwr control	G89 M		66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	off	off	
	Dynamic assist gas	n/a		off	off	off	off	off	off	off	off	off	off	off	
	Assist gas type	G89 A		12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	
	Assist gas press (psi)	G89 I		70*	70*	70*	70*	70*	70*	70*	70*	70*	70*	70*	
	Optional pressure (psi)	G102 I		140	150	160	170	175	180	200	225	225	200	200	
	Power burst time (sec)	G102 T		0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.10	0.10	
DPC	DPC feedrate (ipm)	G89 U		450	420	390	320	290	260	200	160	135	n/a	n/a	
	DPC min power (%)	G89 V		25	25	30	40	40	40	40	40	40	n/a	n/a	
DAG	Dyn assist, near (psi)	G102 A		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
	Dyn assist, far (psi)	G102 B		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
Misc	Focusing lens (inch)	n/a		5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	7.5	7.5	
	Nozzle	n/a		060hc	060hc	060hc	060hc	080hc	080hc	080hc	080hc	080hc	100str	100str	
	Near Field Focus (inch)	n/a		-0.015	-0.015	-0.020	-0.030	-0.035	-0.040	-0.065	-0.080	-0.090	-0.140	-0.155	
	Far Field Focus (inch)	n/a		-0.015	-0.015	-0.020	-0.030	-0.035	-0.040	-0.065	-0.080	-0.090	-0.140	-0.155	

MILD STEEL

O_2 ASSIST, DRY

	Thickness (inch)				**0.048**	**0.060**	**0.075**	**0.090**	**0.105**	**0.120**	**0.135**	**0.188**	**0.250**		
	Thickness (gage)				18	16	14	13	12	11	10	3/16	1/4		
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q			2700	2700	2700	3000	3000	3000	3000	3000	3000		
	Frequency (Hz)	G89 R			200	200	200	200	200	200	150	150	150		
	Duty cycle (%)	G89 R			20	20	20	20	20	20	15	15	12		
	Standoff (inch)	G102 S			0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080		
	Pierce time (sec)	G89 D			**0.00**	**0.00**	**0.10**	**0.10**	**0.15**	**0.2/0.1**	**1.0/0.1**	**1.5/0.15**	**2.0/0.2**		
	Z hold distance (inch)				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	Part coolant (on/off)	G89 J			9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off		
	Ramped pierce (on/off)	G103			off	off	off	off	off	off	on	on	on		
	Assist gas port	G89 B			11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2		
	Assist gas press (psi)	G89 E			20	20	20	20	20	20	20	20	20		
Rapid Pierce	Laser power (Watt)	n/a			n/a	n/a	n/a	n/a	n/a	4000	4000	4000	4000		
	Standoff (inch)	n/a			n/a	n/a	n/a	n/a	n/a	0.200	0.200	0.200	0.200		
	Pierce time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	0.10	0.10	0.15	0.20		
	Cool time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	0.00	0.00	0.00	0.00		
	Airblast off time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	0.10	0.10	0.15	0.20		
	Airblast on time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	0.00	0.00	0.00	0.00		
	Assist gas press (psi)	n/a			n/a	n/a	n/a	n/a	n/a	8	8	8	8		
Ramped Pierce	Number of steps	G103 Q			n/a	n/a	n/a	n/a	n/a	n/a	1	1	2		
	Step 0 power (% max)	G103 R			n/a	n/a	n/a	n/a	n/a	n/a	25	25	25		
	Step 1 time (sec)	G103 A			n/a	n/a	n/a	n/a	n/a	n/a	1.00	1.50	1.50		
	Step 1 power (% max)	G103 S			n/a	n/a	n/a	n/a	n/a	n/a	100	100	100		
	Step 2 time (sec)	G103 B			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.50		
	Step 2 power (% max)	G103 T			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	100		
	Step 3 time (sec)	G103 C			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 3 power (% max)	G103 U			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Cooling time (sec)	G103 F			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
Cutting	Laser power (Watt)	G89 T			1300	1300	150	1500	1500	1750	2000	4000	4000		
	Frequency (Hz)	G89 S			500	500	500	500	500	500	500	500	500		
	Duty cycle (%)	G89 S			95	95	95	95	95	95	95	100	100		
	Pre-cut dwell (sec)	G102 D			0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.10	0.10		
	Feedrate (ipm)	F #148			**275**	**250**	**225**	**175**	**155**	**145**	**135**	**115**	**105**		
	Kerf width (inch)	G89 K			0.009	0.009	0.009	0.009	0.010	0.010	0.011	0.020	0.019		
	Standoff (inch) [NC]	G102 Z			0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.030	0.030		
	Optional standoff (inch)	M45			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Coolant (on/off)	G89 C			9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off		
	Dynamic pwr control	G89 M			66=yes	66=yes	66=yes	66=yes	66=yes	66=yes	66=yes	66=yes	66=yes		
	Dynamic assist gas	n/a			off	off	off	off	off	off	off	off	off		
	Assist gas type	G89 A			11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2		
	Assist gas press (psi)	G89 I			40	40	40	40	40	40	40	7	9		
	Optional pressure (psi)	G102 I			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Power burst time (sec)	G102 T			0.05	0.05	0.00	0.00	0.00	0.00	0.l00	0.05	0.05		
DPC	DPC feedrate (ipm)	G89 U			225	200	180	145	125	115	100	n/a	n/a		
	DPC min power (%)	G89 V			40	40	40	40	40	40	40	n/a	n/a		
DAG	Dyn assist, near (psi)	G102 A			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Dyn assist, far (psi)	G102 B			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
Misc	Focusing lens (inch)	n/a			5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0		
	Nozzle	n/a			060hc	060hc	060hc	060hc	060hc	060hc	060hc	0.040"	0.040"		
	Focus (inch) [From Tip]	n/a			-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	+0.100	+0.085		

MILD STEEL – (COLD-ROLLED OR HRPO)

O2 ASSIST, DRY

	Thickness (inch)				**0.313**	**0.375**	**0.500**	**0.625**	**0.750**	**1.000**					
	Thickness (gage)				5/16	3/8	1/2	5/8	3/4						
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q			3000	3000	4000	4000	4000	4000					
	Frequency (Hz)	G89 R			100	100	100	100	100	100					
	Duty cycle (%)	G89 R			15	15	15	15	15	18					
	Standoff (inch)	G102 S			0.100	0.100	0.100	0.100	0.100	0.100					
	Pierce time (sec)	G89 D			**3.0/0.6**	**4.0/0.8**	**9.5/1.0**	**13.5/1.4**	**22.5/2.5**	**42.0/3.0**					
	Z hold distance (inch)				0.0	0.0	0.0	0.0	0.0	0.0					
	Part coolant (on/off)	G89 J			9=off	9=off	9=off	9=off	9=off	9=off					
	Ramped pierce (on/off)	G103			on	on	on	on	on	on					
	Assist gas port	G89 B			11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2					
	Assist gas press (psi)	G89 E			20	20	20	20	20	20					
Rapid Pierce	Laser power (Watt)	n/a			4000	4000	4000	4000	4000	4000					
	Standoff (inch)	n/a			0.200	0.200	0.200	0.200	0.200	0.200					
	Pierce time (sec)	n/a			0.40	0.60	0.80	1.00	2.00	2.50					
	Cool time (sec)	n/a			0.20	0.20	0.20	0.40	0.50	0.50					
	Airblast off time (sec)	n/a			0.40	0.60	0.80	1.00	2.00	2.50					
	Airblast on time (sec)	n/a			0.20	0.20	0.20	0.40	0.50	0.50					
	Assist gas press (psi)	n/a			8	8	8	8	8	8					
Ramped Pierce	Number of steps	G103 Q			2	2	2	2	2	2					
	Step 0 power (% max)	G103 R			50	50	50	50	50	50					
	Step 1 time (sec)	G103 A			2.5	3.0	6.0	8.0	11.0	20.0					
	Step 1 power (% max)	G103 S			100	100	100	100	100	100					
	Step 2 time (sec)	G103 B			0.5	1.0	3.0	4.5	10.0	20.0					
	Step 2 power (% max)	G103 T			100	100	100	100	100	100					
	Step 3 time (sec)	G103 C			n/a	n/a	n/a	n/a	n/a	n/a					
	Step 3 power (% max)	G103 U			n/a	n/a	n/a	n/a	n/a	n/a					
	Cooling time (sec)	G103 F			n/a	n/a	0.50	1.00	1.50	2.00					
Cutting	Laser power (Watt)	G89 T			4000	4000	4000	4000	4000	4000					
	Frequency (Hz)	G89 S			500	500	500	500	500	500					
	Duty cycle (%)	G89 S			100	100	100	100	85	85					
	Pre-cut dwell (sec)	G102 D			0.10	0.10	0.15	0.20	0.50	0.50					
	Feedrate (ipm)	F #148			**90**	**80**	**55**	**45**	**34**	**22**					
	Kerf width (inch)	G89 K			0.018	0.017	0.018	0.020	0.022	0.024					
	Standoff (inch) [NC]	G102 Z			0.030	0.030	0.040	0.040	0.040	0.040					
	Optional standoff (inch)	M45			n/a	n/a	n/a	n/a	n/a	n/a					
	Coolant (on/off)	G89 C			9=off	9=off	9=off	9=off	9=off	9=off					
	Dynamic pwr control	G89 M			no	no	yes	no	no	no					
	Dynamic assist gas	n/a			off	off	off	off	off	off					
	Assist gas type	G89 A			11=O_2	11=O_2	11=O_2	11=O_2	11=O_2	11=O_2					
	Assist gas press (psi)	G89 I			10	11	8	8	10	10					
	Optional pressure (psi)	G102 I			n/a	n/a	n/a	n/a	n/a	n/a					
	Power burst time (sec)	G102 T			0.10	0.10	0.10	0.10	0.10	0.10					
DPC	DPC feedrate (ipm)	G89 U			n/a	n/a	48	n/a	n/a	n/a					
	DPC min power (%)	G89 V			n/a	n/a	35	n/a	n/a	n/a					
DAG	Dyn assist, near (psi)	G102 A			n/a	n/a	n/a	n/a	n/a	n/a					
	Dyn assist, far (psi)	G102 B			n/a	n/a	n/a	n/a	n/a	n/a					
Misc	Focusing lens (inch)	n/a			7.5	7.5	7.5	7.5	7.5	7.5					
	Nozzle	n/a			0.040"	0.040"	062shr	062shr	062shr	062shr					
	Near Field Focus (inch)	n/a			-0.025	-0.025	-0.020	0.000	+0.020	+0.020					
	Far Field Focus (inch)	n/a			-0.025	-0.025	-0.020	0.000	+0.020	+0.020					

STAINLESS STEEL (TYPE 304)

N_2 ASSIST

	Thickness (inch)				**0.030**	**0.036**	**0.048**	**0.060**	**0.075**	**0.090**	**0.105**	**0.120**	**0.135**		
	Thickness (gage)				22	20	18	16	14	13	12	11	10		
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q			2700	2700	2700	3000	3000	3000	3000	3000	3000		
	Frequency (Hz)	G89 R			200	200	200	200	200	200	200	200	500		
	Duty cycle (%)	G89 R			20	20	20	20	20	20	20	20	100		
	Standoff (inch)	G102 S			0.080	0.080	0.080	0.080	0.080	0.080	0.100	0.100	0.100		
	Pierce time (sec)	G89 D			**0.00**	**0.00**	**0.00**	**0.01**	**0.01**	**0.05**	**0.10**	**0.15**	**0.20**		
	Z hold distance (inch)				0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1		
	Part coolant (on/off)	G89 J			9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off		
	Ramped pierce (on/off)	G103			off	off	off	off	off	off	off	off	off		
	Assist gas port	G89 B			12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2		
	Assist gas press (psi)	G89 E			30	30	30	30	30	30	30	30	30		
Rapid Pierce	Laser power (Watt)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Standoff (inch)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Pierce time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Cool time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Airblast off time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Airblast on time (sec)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Assist gas press (psi)	n/a			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
Ramped Pierce	Number of steps	G103 Q			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 0 power (% max)	G103 R			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 1 time (sec)	G103 A			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 1 power (% max)	G103 S			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 2 time (sec)	G103 B			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 2 power (% max)	G103 T			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 3 time (sec)	G103 C			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Step 3 power (% max)	G103 U			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
	Cooling time (sec)	G103 F			n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a		
Cutting	Laser power (Watt)	G89 T			4000	4000	4000	4000	4000	4000	4000	4000	4000		
	Frequency (Hz)	G89 S			2000	2000	2000	2000	2000	2000	2000	2000	2000		
	Duty cycle (%)	G89 S			100	100	100	100	100	100	100	100	100		
	Pre-cut dwell (sec)	G102 D			0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		
	Feedrate (ipm)	F #148			**400**	**375**	**335**	**305**	**250**	**220**	**195**	**170**	**150**		
	Kerf width (inch)	G89 K			0.008	0.008	0.009	0.009	0.009	0.009	0.009	0.009	0.010		
	Standoff (inch) [NC]	G102 Z			0.025	0.025	0.025	0.025	0.025	0.100*	0.100*	0.100*	0.100*		
	Optional standoff (inch)	M45			n/a	n/a	n/a	n/a	n/a	0.025	0.025	0.025	0.025		
	Coolant (on/off)	G89 C			9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off		
	Dynamic pwr control	G89 M			66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on		
	Dynamic assist gas	n/a			off	off	on	on	on	on	on	on	on		
	Assist gas type	G89 A			12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2		
	Assist gas press (psi)	G89 I			100	100	70*	70*	70*	70*	70*	70*	70*		
	Optional pressure (psi)	G102 I			n/a	n/a	130	150	150	150	160	160	170		
	Power burst time (sec)	G102 T			0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05		
DPC	DPC feedrate (ipm)	G89 U			340	325	270	250	200	220	195	170	150		
	DPC min power (%)	G89 V			20	20	20	20	15	15	15	15	15		
DAG	Dyn assist, near (psi)	G102 A			n/a	n/a	170	190	170	170	175	175	185		
	Dyn assist, far (psi)	G102 B			n/a	n/a	130	150	150	150	160	160	170		
Misc	Focusing lens (inch)	n/a			5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0		
	Nozzle	n/a			060hc	060hc	080hc	080hc	080hc	080hc	080hc	080hc	080hc		
	Near Field Focus (inch)	n/a			-0.020	-0.020	-0.020	-0.020	-0.020	-0.020	-0.030	-0.040	-0.050		
	Far Field Focus (inch)	n/a			-0.020	-0.020	-0.020	-0.020	-0.020	-0.020	-0.030	-0.040	-0.050		

STAINLESS STEEL (TYPE 304)

N_2 ASSIST

	Thickness (inch)				**0.188**	**0.250**	**0.313**	**0.375**	**0.500**						
	Thickness (gage)				3/16	1/4	5/16	3/8	1/2						
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q			4000	4000	4000	4000	4000						
	Frequency (Hz)	G89 R			500	500	500	500	125						
	Duty cycle (%)	G89 R			100	100	100	100	75						
	Standoff (inch)	G102 S			0.100	0.100	0.100	0.125	0.200						
	Pierce time (sec)	G89 D			**0.35**	**0.50**	**0.75**	**1.25**	**7.00**						
	Z hold distance (inch)				0.1	0.2	0.2	0.3	0.3						
	Part coolant (on/off)	G89 J			9=off	9=off	9=off	9=off	9=off						
	Ramped pierce (on/off)	G103			off	off	off	off	off						
	Assist gas port	G89 B			12=N_2	12=N_2	12=N_2	12=N_2	12=N_2						
	Assist gas press (psi)	G89 E			30	30	30	30	12						
Rapid Pierce	Laser power (Watt)	n/a			n/a	n/a	n/a	n/a	n/a						
	Standoff (inch)	n/a			n/a	n/a	n/a	n/a	n/a						
	Pierce time (sec)	n/a			n/a	n/a	n/a	n/a	n/a						
	Cool time (sec)	n/a			n/a	n/a	n/a	n/a	n/a						
	Airblast off time (sec)	n/a			n/a	n/a	n/a	n/a	n/a						
	Airblast on time (sec)	n/a			n/a	n/a	n/a	n/a	n/a						
	Assist gas press (psi)	n/a			n/a	n/a	n/a	n/a	n/a						
Ramped Pierce	Number of steps	G103 Q			n/a	n/a	n/a	n/a	n/a						
	Step 0 power (% max)	G103 R			n/a	n/a	n/a	n/a	n/a						
	Step 1 time (sec)	G103 A			n/a	n/a	n/a	n/a	n/a						
	Step 1 power (% max)	G103 S			n/a	n/a	n/a	n/a	n/a						
	Step 2 time (sec)	G103 B			n/a	n/a	n/a	n/a	n/a						
	Step 2 power (% max)	G103 T			n/a	n/a	n/a	n/a	n/a						
	Step 3 time (sec)	G103 C			n/a	n/a	n/a	n/a	n/a						
	Step 3 power (% max)	G103 U			n/a	n/a	n/a	n/a	n/a						
	Cooling time (sec)	G103 F			n/a	n/a	n/a	n/a	n/a						
Cutting	Laser power (Watt)	G89 T			4000	4000	4000	4000	4000						
	Frequency (Hz)	G89 S			2000	2000	2000	2000	2000						
	Duty cycle (%)	G89 S			100	100	100	100	100						
	Pre-cut dwell (sec)	G102 D			0.00	0.00	0.00	0.00	0.00						
	Feedrate (ipm)	F #148			**100**	**75**	**55**	**42**	**12**						
	Kerf width (inch)	G89 K			0.011	0.012	0.013	0.015	0.017						
	Standoff (inch) [NC]	G102 Z			0.100*	0.100*	0.100*	0.100*	0.100*						
	Optional standoff (inch)	M45			0.025	0.025	0.025	0.025	0.025						
	Coolant (on/off)	G89 C			9=off	9=off	9=off	9=off	9=off						
	Dynamic pwr control	G89 M			66=on	66=on	66=on	66=on	66=on						
	Dynamic assist gas	n/a			off	off	off	off	off						
	Assist gas type	G89 A			12=N_2	12=N_2	12=N_2	12=N_2	12=N_2						
	Assist gas press (psi)	G89 I			70*	70*	70*	30*	30*						
	Optional pressure (psi)	G102 I			180	200	230	230	230						
	Power burst time (sec)	G102 T			0.10	0.10	0.10	0.10	0.10						
DPC	DPC feedrate (ipm)	G89 U			100	75	50	n/a	n/a						
	DPC min power (%)	G89 V			55	55	55	n/a	n/a						
DAG	Dyn assist, near (psi)	G102 A			n/a	n/a	n/a	n/a	n/a						
	Dyn assist, far (psi)	G102 B			n/a	n/a	n/a	n/a	n/a						
Misc	Focusing lens (inch)	n/a			7.5	7.5	7.5	7.5	7.5						
	Nozzle	n/a			100str	100str	125sp	125sp	125sp						
	Near Field Focus (inch)	n/a			-0.135	-0.150	-0.100	-0.275	-0.400						
	Far Field Focus (inch)	n/a			-0.135	-0.150	-0.100	-0.275	-0.400						

ALUMINUM (5052-H32)

N_2 ASSIST

	Thickness (inch)		**0.020**	**0.032**	**0.050**	**0.063**	**0.080**	**0.090**	**0.100**	**0.125**	**0.190**	**0.250**	**0.313**	**0.375**	**0.500**
	Thickness (gage)									1/8	3/16	1/4	5/16	3/8	1/2
	Parameters	Code													
Pierce	Laser power (Watt)	G89 Q	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000
	Frequency (Hz)	G89 R	500	500	500	500	500	500	500	500	35	35	35	35	50
	Duty cycle (%)	G89 R	100	100	100	100	100	100	100	100	70	70	70	70	70
	Standoff (inch)	G102 S	0.080	0.080	0.080	0.080	0.080	0.080	0.100	0.100	0.100	0.100	0.100	0.100	0.200
	Pierce time (sec)	G89 D	**0.00**	**0.00**	**0.00**	**0.00**	**0.01**	**0.01**	**0.01**	**0.10**	**1.00**	**1.50**	**3.00**	**4.50**	**7.00**
	Z hold distance (inch)		0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2
	Part coolant (on/off)	G89 J	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off
	Ramped pierce (on/off)	G103	off	off	off	off	off	off	off	off	off	off	off	off	off
	Assist gas port	G89 B			12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2
	Assist gas press (psi)	G89 E	30	30	30	30	30	30	30	30	30	30	30	30	10
Rapid Pierce	Laser power (Watt)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Standoff (inch)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Pierce time (sec)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Cool time (sec)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Airblast off time (sec)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Airblast on time (sec)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Assist gas press (psi)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Ramped Pierce	Number of steps	G103 Q	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 0 power (% max)	G103 R	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 1 time (sec)	G103 A	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 1 power (% max)	G103 S	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 2 time (sec)	G103 B	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 2 power (% max)	G103 T	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 3 time (sec)	G103 C	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Step 3 power (% max)	G103 U	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Cooling time (sec)	G103 F	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Cutting	Laser power (Watt)	G89 T	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000	4000
	Frequency (Hz)	G89 S	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000
	Duty cycle (%)	G89 S	100	100	100	100	100	100	100	100	100	100	100	100	100
	Pre-cut dwell (sec)	G102 D	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Feedrate (ipm)	F #148	**600**	**525**	**475**	**400**	**315**	**300**	**250**	**170**	**80**	**70**	**42**	**35**	**22**
	Kerf width (inch)	G89 K	0.007	0.007	0.008	0.008	0.008	0.008	0.009	0.009	0.010	0.010	0.010	0.010	0.010
	Standoff (inch) [NC]	G102 Z	0.025	0.025	0.025	0.025	0.025	0.100*	0.100*	0.100*	0.100*	0.100*	0.100*	0.100*	0.100*
	Optional standoff (inch)	M45	n/a	n/a	n/a	n/a	n/a	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025
	Coolant (on/off)	G89 C	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off	9=off
	Dynamic pwr control	G89 M	66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	66=on	off	off	off
	Dynamic assist gas	n/a	off	of	off	off	off	of	off	off	off	off	off	off	off
	Assist gas type	G89 A	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2	12=N_2
	Assist gas press (psi)	G89 I	50	60	70	85	30*	30*	30*	30*	30*	30*	30*	30*	30*
	Optional pressure (psi)	G102 I	n/a	n/a	n/a	n/a	90	90	110	110	150	150	150	180	180
	Power burst time (sec)	G102 T	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.10	0.10	0.10	0.10	0.10
DPC	DPC feedrate (ipm)	G89 U	540	475	425	360	285	270	225	155	72	63	n/a	n/a	n/a
	DPC min power (%)	G89 V	30	30	30	30	30	20	20	20	75	75	n/a	n/a	n/a
DAG	Dyn assist, near (psi)	G102 A	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Dyn assist, far (psi)	G102 B	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Misc	Focusing lens (inch)	n/a	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	7.5	7.5	7.5	7.5	7.5
	Nozzle	n/a	060hc	060hc	060hc	080hc	080hc	080hc	080hc	080hc	080hc	125sp	125sp	125sp	125sp
	Near Field Focus (inch)	n/a	-0.045	-0.050	-0.055	-0.055	-0.065	-0.060	-0.055	-0.050	-0.125	-0.070	-0.100	-0.150	-0.210
	Far Field Focus (inch)	n/a	-0.045	-0.050	-0.055	-0.055	-0.065	-0.060	-0.055	-0.050	-0.125	-0.070	-0.100	-0.150	-0.210

MILD STEEL

N_2 ASSIST

Material	MS030N2	MS036N2	MS048N2	MS060N2	MS075N2	MS090N2	MS105N2	MS120N2	MS135N2	MS188N2	MS250N2
Process	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut
Lead-in Length (inch)	Approx. 0.125	Approx. 0.125	Approx. 0.125	Approx. 0.125	Approx. 0.125	Approx. 0.125	Approx. 0.125	Approx. 0.200	Approx. 0.200	Approx. 0.200	Approx. 0.200
Lead-in Angle	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°
Lead-in Feed	Approx. 30%	Approx. 30%	Approx. 30%	Approx. 30%	Approx. 30%	Approx. 30%	Approx. 30%	Approx. 20%	Approx. 20%	Approx. 20%	Approx. 10%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners
Optional Gas (M67)	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.
Ramped Start/End	No	No	No	No	No	No	No	No	No	No	No
Corner Ramping	No	No	No	No	No	No	No	No	No	No	No
Start Cut	G84	G84	G84	G84	G84	G84	G84	G84	G84	G84	G84
Optional Z Standoff (M45)	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130

MILD STEEL

O_2 ASSIST

Material	MS048O2	MS060O2	MS075O2	MS090O2	MS105O2	MS120O2	MS135O2	MS188O2	MS250O2
Process	Cut	Cut	Cut	Cut	Cut	Cut	Cut	Cut	Cut
Lead-in Length (inch)	~ 0.050	~ 0.050	~ 0.050	~ 0.100	~ 0.125	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050
						G84 T2 ~ 0.100	G84 T2 ~ 0.125	G84 T2 ~ 0.150	G84 T2 ~ 0.180
Lead-in Angle	45° to 75°	45° to 75°	45° to 75°	45° to 75°	45° to 75°	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°
						G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°
Lead-in Feed	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	No	No	No	No	No	No	No	No	No
Optional Gas (M67)	No	No	No	No	No	No	No	No	No
Ramped Start/End	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%
Corner Ramping	No	No	No	No	No	No	No	No	No
Start Cut	G84	G84	G84	G84	G84	G84 or G84T2	G84 or G84T2	G84 or G84T2	G84 or G84T2
Optional Z Standoff (M45)	No	No	No	No	No	No	No	No	No
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130

MILD STEEL

O_2 ASSIST (CONTINUED)

Material	MS313O2	MS375O2	MS500O2	MS625O2	MS750O2	MS1000O2
Process	Cut	Cut	Cut	Cut	Cut	Cut
Lead-in Length (inch)	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050	G84 ~ 0.050
	G84 T2 ~ 0.200	G84 T2 ~ 0.200	G84 T2 ~ 0.300	G84 T2 ~ 0.300	G84 T2 ~ 0.300	G84 T2 ~ 0.400
Lead-in Angle	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°	G84 45° to 75°
	G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°	G84 T2 45° to 75°
Lead-in Feed	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%	~ 60%
Lead-in Exact Stop	No	No	No	No	No	No
Exact Stop (features)	No	No	No	No	No	No
Optional Gas (M67)	No	No	No	No	No	No
Ramped Start/End	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	2 segments both~0.075" Start 60/80% End 80/60%	1 segment 0.125" End 80%	1 segment 0.125" End 80%
Corner Ramping	No	No	No	No	No	No
Start Cut	G84 or G84T2	G84 or G84T2	G84 or G84T2	G84 or G84T2	G84 or G84T2	G84 or G84T2
Optional Z Standoff (M45)	No	No	No	No	No	No
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130

STAINLESS STEEL (TYPE 304)

N_2 ASSIST

Material	SS030N2	SS036N2	SS048N2	SS060N2	SS075N2	SS090N2	SS105N2	SS120N2	SS135N2
Process	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut
Lead-in Length (inch)	~ 0.050	~ 0.050	~ 0.050	~ 0.050	~ 0.050	~ 0.050	~ 0.125	~ 0.150	~ 0.150
Lead-in Angle	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°
Lead-in Feed	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 20%	~ 20%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners
Optional Gas (M67)	No	No	Yes; + add dwell for 0.200 sec.	Yes; + add dwell for 0.200 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.
Ramped Start/End	No	No	No	No	No	No	No	No	No
Corner Ramping	No	No	No	No	No	No	No	No	No
Start Cut	G84	G84	G84	G84	G84	G84	G84	G84	G84
Optional Z Standoff (M45)	No	No	No	No	No	Yes	Yes	Yes	Yes
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130

STAINLESS STEEL (TYPE 304)

N_2 ASSIST (CONTINUED)

Material	SS188N2	SS250N2	SS313N2	SS375N2	SS500N2
Process	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut
Lead-in Length (inch)	~ 0.200	~ 0.200	~ 0.250	~ 0.300	~ 0.350
Lead-in Angle	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°
Lead-in Feed	~ 20%	~ 10%	~ 30%	~ 30%	~ 30%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	Yes, for sharp corners	Yes, for sharp corners	No	No	No
Optional Gas (M67)	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.
Ramped Start/End	No	No	No	3 segments all 0.075" Start 60/70/80%	3 segments all 0.075" Start 60/70/80%
Corner Ramping	No	No	No	3 segments all 0.020" 60/70/80%	3 segments all 0.020" 60/70/80%
Start Cut	G84	G84	G84	G84	G84
Optional Z Standoff (M45)	Yes	Yes	Yes	Yes	Yes
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130

ALUMINUM (5052-H32)

N_2 ASSIST

Material	AL020N2	AL032N2	SS050N2	AL063N2	AL080N2	AL090N2	AL100N2	AL125N2
Process	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut
Lead-in Length (inch)	~ 0.050	~ 0.050	~ 0.050	~ 0.080	~ 0.100	~ 0.100	~ 0.150	~ 0.150
Lead-in Angle	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°	45° to 90°
Lead-in Feed	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 30%	~ 20%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners	Yes, for sharp corners
Optional Gas (M67)	No	No	No	No	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.300 sec.
Ramped Start/End	No	No	No	No	No	No	No	No
Corner Ramping	No	No	No	No	No	No	No	No
Start Cut	G84	G84	G84	G84	G84	G84	G84	G84
Optional Z Standoff (M45)	No	No	No	No	No	Yes	Yes	Yes
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130	Use M130

ALUMINUM (5052-H32)

N_2 ASSIST (CONTINUED)

Material	AL190N2	AL250N2	AL313N2	AL375N2	AL500N2
Process	Clean Cut	Clean Cut	Clean Cut	Clean Cut	Clean Cut
Lead-in Length (inch)	~ 0.200	~ 0.250	~ 0.300	~ 0.350	~ 0.400
Lead-in Angle	60° to 90°	60° to 90°	60° to 90°	60° to 90°	60° to 90°
Lead-in Feed	~ 20%	~ 20%	~ 20%	~ 10%	~ 10%
Lead-in Exact Stop	Yes	Yes	Yes	Yes	Yes
Exact Stop (features)	Yes, for sharp corners	Yes, for sharp corners	No	No	No
Optional Gas (M67)	Yes; + add dwell for 0.300 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.	Yes; + add dwell for 0.500 sec.
Ramped Start/End	No	No	No	3 segments all 0.030" Start 30/50/80%	3 segments all 0.030" Start 30/50/80%
Corner Ramping	No	No	No	3 segments all 0.020" 60/70/80%	3 segments all 0.020" 60/70/80%
Start Cut	G84	G84	G84	G84	G84
Optional Z Standoff (M45)	Yes	Yes	Yes	Yes	Yes
Z-axis Antidive	Use M130	Use M130	Use M130	Use M130	Use M130

SECTION D DIAGNOSTIC GUIDE

GENERAL DIAGNOSTIC GUIDELINES

HISTORY

Find out what happened since it last worked correctly. Even those events that seem to be insignificant often help identify the cause of the problem. Talk to other operators, programmers, and maintenance personnel.

PATTERNS

Look for patterns in the problem. Does the problem occur at a certain time of day or after a certain period of cutting time? Is it related to position on the worktable, orientation of the part, size of cutout, upper/lower pallet, personnel, or material condition? Does the problem occur at certain stages of the cutting process such as pierce, lead-in, or contour?

SETTINGS

Verify that all cutting parameters and machine adjustments are correct. Refer to the parameter tables for recommended parameters. Refer to the process notes for hints and suggestions. Check the following machine adjustments as described in the Operation, Safety, and Maintenance manual:

1. Nozzle tip is clean and undamaged.
2. Standoff is correct.
3. Focusing lens is correct focal length, in good condition, and mounted correctly.
4. Lens is centered.
5. Focus setting is correct.

DIAGNOSIS

Use the diagnostic charts for the type of material you are cutting to identify possible causes of the problem and recommended remedies. Charts are provided for the following materials:

- Aluminum
- Mild Steel and Tool Steel (Oxygen Assist)
- Mild Steel, Galvanized, and Stainless Steel (N_2 Assist)
- Copper Alloys
- Composites & Plastics
- Wood

DIAGNOSTIC CHARTS (OVERVIEW)

Each chart contains three distinct categories of cutting problems - (1) pierce problems, (2) edge condition problems that are independent of location along the edge, and (3) edge condition problems that occur at certain locations or for certain kinds of cutouts. Use the charts as follows:

- Identify the problem category (pierce, edge condition, etc.).
- Locate the problem description in the “problem” column that best matches your cutting problem.
- Identify possible causes of this problem. The possible causes are listed to the right of each problem in the “possible causes” column. The most likely causes are listed first.
- Identify remedies for the problem. The remedies for each cause are listed to the right of the cause in the “remedy” column. The most common remedies are listed first.
- Starting with the first recommended remedy, make the necessary change to the cutting process. A combination of these remedies may be required to correct the cutting problem(s).

APPLICATIONS SUPPORT

If you are unable to identify and resolve the problem yourself, use the worksheet in the back of this section to document specifics of the problem. Then contact the Laser Applications Department at CINCINNATI INCORPORATED for additional assistance.

ALUMINUM

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
No sparks during the pierce.	• Standoff too low.	• Increase standoff.
Bright plasma during the pierce	• Standoff too low. • Assist gas pressure too high.	• Increase standoff. • Reduce assist gas pressure.
Blowout at the end of a pierce just as cutting head starts to move.	• Pierce not complete. Test: observe pierce process.	• Increase pierce power (duty cycle). • Increase pierce time.
Tip touch alarm at the end of a pierce just as cutting head starts to move.	• Pierce mound buildup has come in contact with the nozzle tip.	• Use optional standoff (M45). • Use anti-spatter spray on material before cutting.
Edge quality problems (independent of location along part edge)		
Loud popping noises during cutting and gouges in the part edge.	• Assist gas pressure too high (if using air or oxygen). • Feedrate too high.	• Reduce assist gas pressure. • Reduce feedrate.
Dross. Note: For thick aluminum, dross is typical.	• Focus not optimal. • Assist gas pressure too low. • Feed rate too high.	• Adjust focus (approx. 0.020" (.5mm)) and observe effect. • Increase assist gas pressure (approx. 10%). • Reduce feed rate (approx. 10%).
Edge quality problems (depends on location along part edge or size of cutout)		
Dross on one side or two opposite sides of a part.	• Lens not centered. • Nozzle partially clogged or damaged. • Lens dirty or damaged. Test: cut with a known good lens. • Beam delivery system out of alignment. Test: check beam alignment. • Laser mode asymmetric. Test: check mode quality.	• Center the lens. • Clean or replace nozzle. • Clean or replace lens. • Call maintenance to align beam delivery system. • Call maintenance to correct the mode problem.
Very rough edge or lost cut at beginning of part edge (0.5" to 2" (13 mm to 51 mm)). Usually accompanied by loud "popping" noise.	• Pierce not complete. Test: observe pierce process. • Start-of-cut feed rate and assist gas pressure too high. • Pre-cut dwell incorrect. • Dynamic power minimum too low.	• See remedy for blowout at end of pierce. • Use start-of-cut technique (reduced assist gas pressure, reduced feed rate). • Adjust pre-cut dwell time. • Increase dynamic power minimum.
Loss of cut or dross at corners or on intricate features.	• Power too low at reduced federate.	• Increase dynamic power minimum.

MILD STEEL AND TOOL STEEL – O_2 ASSIST

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
Blowout at the beginning of the pierce.	• Power too high at start of pierce. • Assist gas pressure too high.	• Reduce start % of ramped pierce power. • Reduce assist gas pressure (approx. 10%)
Blowout during the pierce.	• Power too high too fast during pierce (for thick material using ramped pierce). • Assist gas pressure too high.	• Reduce starting ramped duty %. • Reduce ending ramped duty %. • Increase ramped pierce time. • Reduce assist gas pressure (approx. 10%).
Blowout at the end of a pierce just as cutting head starts to move.	• Pierce not complete. Test: observe pierce process.	• Increase pierce time. • Increase ramped hold time (ramped pierce only). • Increase pierce power.
Blowout in thin material.	• Pierce not complete. • Power too high.	• Increase pierce time. • Increase power. • Reduce power. • Reduce assist gas pressure (approx 10%).
Pierce Problems – Rapid Pierce		
Blowout at the end of a pierce just as cutting head starts to move.	• Pierce not complete. Test: observe pierce process.	• Lower standoff. • Increase pierce time. • Increase assist gas pressure.
Pierce hole elongated at top.	• Air blast is distorting pierce hole.	• Increase air blast off-time.
Poor pierce mound removal	• Pierce mound cools before air blast is activated • Air blast is not strong enough. • Non-oiled material	• Reduce air blast off-time. • Increase air blast pressure. • Move air blast nozzle closer to workpiece. • Use auto-mist. • Increase auto-mist time or flow rate.

MILD STEEL AND TOOL STEEL – O_2 ASSIST (CONT.)

Edge quality problems (independent of location along part edge)		
Very rough part edge. Wide kerf.	• Too hot, thermal runaway.	• Reduce assist gas pressure (approx. 10%). • Adjust focus (approx 0.020" (0.5mm)) and observe effect. • Reduce power (approx. 5%). • Increase feedrate (approx. 5%) • Increase pierce cooling time. • Use ramped feedrate start of cut.
	• Poor work material or material surface.	• Use clean, scale-free, gouge-free work material.
Small gouges widely spaced. Typical gouge extends from the middle to the bottom of the part edge.	• Assist gas pressure slightly high. • Feedrate too slow. • Deformities in work material or material surface.	• Reduce pressure (approx. 5%). • Increase feedrate. • Use clean, scale-free, gouge-free work material.
Dross.	• Focus not optimal.	• Adjust focus (approx. 0.020" (.5mm)) and observe effect.
	• Assist gas pressure too low.	• Increase pressure (approx. 10%).
	• Power too low.	• Increase power (approx. 5%).
	• Feed rate too high.	• Reduce feed rate (approx. 5%).
Edge quality problems (depends on location along part edge or size of cutout)		
Very rough edge or lost cut at beginning of part edge (0.5" to 2" (13 mm to 51 mm)).	• Pierce not complete. Test: observe pierce process.	• See remedy for blowout at end of pierce (pulse pierce or rapid pierce).
	• Lead-in feedrate too high.	• Lead-in feedrate should be 70% to 100% of contour feedrate.
	• Lead-in transition too sharp.	• Use angled lead-in or radius the transition from lead-in to part edge. • Use a ramped start of cut.
	• Pre-cut dwell time incorrect.	• Adjust pre-cut dwell time (0.050 sec increments).
	• Power burst time not set.	• Set power burst time to recommended value (if available on machine).
	• Poor pierce mound removal (rapid pierce).	• See remedy under pierce problems – rapid pierce.
	• Failed to establish clean kerf near bottom of pierce hole in first half of lead-in.	• Use higher assist gas pressure (about 50% higher) for first half of pierce.
	• Not enough power to start the cut.	• Increase dynamic power minimum.
	• Start of cut is too hot.	• Use a ramped start of cut.

MILD STEEL AND TOOL STEEL – O_2 ASSIST (CONT.)

Dross on one side or two opposite sides of a part.	• Lens not centered. • Nozzle partially clogged or damaged. • Lens dirty or damaged. Test: cut with a known good lens. • Beam delivery system out of alignment. Test: check beam alignment. • Laser mode asymmetric. Test: check mode quality.	• Center the lens. • Clean or replace nozzle. • Clean or replace lens. • Call maintenance to align beam delivery system. • Call maintenance to correct the mode problem.
Dross at corners, in small holes, and/or in first part of cut after pierce.	• Power too low for the cutting speed.	• Increase dynamic power minimum.
Gouges at corners, in small holes, and/or in first part of cut after pierce.	• Power too high for the cutting speed.	• Decrease dynamic power minimum.
Gouges just past the corners.	• Localized overheating of sharp corner.	• Use a radius (10% to 15% of material thickness) at the corner.
Very rough edge or lost cut on small cutouts (such as small diameter holes).	• Feedrate too high. • Overheating of cutout (slug). • Pre-heating of adjacent work material.	• Feedrate for small cutouts should be 70% to 100% of contour feedrate. • Reduce lead-in length. • Use angled lead-in or radius the transition from lead-in to part edge to clean up transition. • Re-sequence closely spaced cutouts to allow cooling time for adjacent cutouts.
Spatter on bottom of lens.	• Pierce frequency is too low. • Pierce standoff is too low. • Pierce assist gas pressure is too low.	• Increase pierce frequency (normal pierce to 200 Hz, ramped pierce to 100-200 Hz. • Increase pierce standoff. • Increase pierce assist gas pressure.

MILD STEEL, GALVANIZED, AND STAINLESS STEEL – N_2 ASSIST

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
Blue plasma formation during pierce (slows material penetration).	• Assist gas pressure too high. • Focus too low.	• Reduce pressure (approx 10%). • Raise the focus (approx 0.020")
Spatter on bottom of lens.	• Standoff too low. • Assist gas pressure too low. • Frequency is too low.	• Increase standoff. • Increase pressure (10%). • Increase frequency to 200 Hz.
Edge quality problems (independent of location along part edge)		
Consistent, long "icicle" dross on part and/or on parent material.	• Assist gas pressure too low. • Feedrate too low. • Focus too high.	• Increase pressure (approx 10%). • Increase feedrate. • Lower the focus (approx 0.020").
Fine dross.	• Assist gas pressure too high. • Feedrate too high. • Focus too low.	• Reduce pressure (approx 10%). • Reduce feedrate. • Raise the focus (approx 0.020")
Loss of cut	• Feedrate too fast. • Assist gas pressure too high. • Focus lens is dirty.	• Decrease feedrate. • Reduce pressure (approx 10%). • Clean focus lens. (Refer to lens cleaning procedure in the *Operation, Safety, and Maintenance manual*.)
Lower portion of cut edge not smooth and uniform.	• Standoff incorrect. • Nozzle orifice diameter incorrect.	• Check standoff calibration and commanded standoff. • Verify proper nozzle and size.
Edge quality problems (depends on location along part edge or size of cutout)		
Dross on one side or two opposite sides of a part.	• Lens not centered. • Nozzle partially clogged or damaged. • Lens dirty or damaged. Test: cut with a known good lens. • Beam delivery system out of alignment. Test: check beam alignment. • Beam delivery mirrors dirty or damaged. Test: check mode quality. • Laser mode asymmetric. Test: check mode quality.	• Center the lens. • Clean or replace nozzle. • Clean or replace lens. • Call maintenance to align beam delivery system. • Call maintenance to inspect mirrors and clean or replace as necessary. • Call maintenance to correct the mode problem.
Very rough edge or lost cut at beginning of part edge (1/2" to 2" (13mm to 51mm)).	• Pierce not complete. Test: observe pierce process. • Start-of-cut feed rate and assist gas pressure too high. • Lead-in length too short.	• See remedy for blowout at end of pierce. • Use start-of-cut technique (reduced pressure, optional assist gas M67, reduced lead-in feed rate). • Use lead-in lengths of 0.050 to 0.200".
Loss of cut or fine dross at corners or on intricate features.	• Power too low at reduced federate.	• Increase dynamic power minimum.
Long "icicle" dross at corners and on intricate features.	• Power too high at reduce feedrate.	• Decrease dynamic power minimum.

COPPER ALLOYS

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
No "sparks" on pierce..	• Beam is reflecting off workpiece.	• Increase standoff. • Reduce assist gas pressure (approx 10%).
Blue plasma formation during pierce.	• Assist gas pressure too high. • Focus too low.	• Reduce pressure (approx 10%). • Raise the focus (approx 0.020")
Spatter on bottom of lens.	• Standoff too low. • Assist gas pressure too low.	• Increase standoff. • Increase pressure (10%).
Edge quality problems (independent of location along part edge)		
Consistent, long "icicle" dross on part and/or on parent material.	• Assist gas pressure too low. • Feedrate too low. • Focus too high.	• Increase pressure (approx 10%). • Increase feedrate. • Lower the focus (approx 0.020").
Fine dross.	• Assist gas pressure too high. • Feedrate too high. • Focus too low.	• Reduce pressure (approx 10%). • Reduce feedrate. • Raise the focus (approx 0.020")
Loss of cut	• Feedrate too fast. • Assist gas pressure too high. • Focus lens is dirty.	• Decrease feedrate. • Reduce pressure (approx 10%). • Clean focus lens. (Refer to lens cleaning procedure in the *Operation, Safety, and Maintenance manual*.)
Lower portion of cut edge not smooth and uniform.	• Standoff incorrect. • Nozzle orifice diameter incorrect.	• Check standoff calibration and commanded standoff. • Verify proper nozzle and size.
Edge quality problems (depends on location along part edge or size of cutout)		
Dross on one side or two opposite sides of a part.	• Lens not centered. • Nozzle partially clogged or damaged. • Lens dirty or damaged. Test: cut with a known good lens. • Beam delivery system out of alignment. Test: check beam alignment. • Beam delivery mirrors dirty or damaged. Test: check mode quality. • Laser mode asymmetric. Test: check mode quality.	• Center the lens. • Clean or replace nozzle. • Clean or replace lens. • Call maintenance to align beam delivery system. • Call maintenance to inspect mirrors and clean or replace as necessary. • Call maintenance to correct the mode problem.
Dross at lead-in tangency, small holes and/or sharp corners.	• Feedrate too slow.	• Increase feedrate.
Loss of cut during lead-in.	• Feedrate too high. • Assist gas pressure too high. • Power too low for start-of-cut. • Pierce not complete.	• Reduce feedrate. • Reduce pressure (approx 10%). • Use power burst (if available). • Increase pierce time.
Loss of cut or fine dross at corners or on intricate features.	• Power too low at reduced federate.	• Increase dynamic power minimum.
Long "icicle" dross at corners or on intricate features.	• Power too high at reduce feedrate.	• Decrease dynamic power minimum.

COMPOSITES AND PLASTICS

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
Loud "popping" sound at the end of the pierce.	• Pierce not complete. • Assist gas pressure change too great from pierce to cut.	• Increase pierce time (approx 10%). • Reduce cut pressure (approx 10%). • Increase pierce pressure (approx 10%).
Large pierce hole.	• Power too high. • Focus not optimal. • Assist gas pressure too high.	• Reduce power. • Adjust focus. • Reduce pierce pressure (approx 10%).
Edge quality problems (independent of location along part edge)		
Gouges in part edge.	• Assist gas pressure too high. • Power too high. • Feedrate too low. • Focus too low.	• Reduce pressure (approx 10%). • Decrease cut power or duty cycle. • Increase feedrate. • Raise focus (approx 0.020").
Rough edge near bottom of cut.	• Power too low. • Assist gas pressure too low. • Nozzle orifice too small.	• Increase cut power or duty cycle. • Increase pressure (approx 10%). • Use larger orifice nozzle.
Edge is tapered. Note: Tapered edge is normal with some plastics.	• Feedrate too low. • Focus too low.	• Increase feedrate. • Raise the focus.
Flames produced under work piece..	• Assist gas pressure too low.	• Increase pressure (approx 10%). • Use only nitrogen assist gas.
Material re-casting onto bottom of part.	• Assist gas pressure too low. • Feedrate too high.	• Increase pressure (approx 10%). • Reduce feedrate.
Edge quality problems (depends on location along part edge or size of cutout)		
Dross on one side or two opposite sides of a part.	• Lens not centered. • Nozzle partially clogged or damaged. • Lens dirty or damaged. Test: cut with a known good lens. • Beam delivery system out of alignment. Test: check beam alignment. • Beam delivery mirrors dirty or damaged. Test: check mode quality. • Laser mode asymmetric. Test: check mode quality.	• Center the lens. • Clean or replace nozzle. • Clean or replace lens. • Call maintenance to align beam delivery system. • Call maintenance to inspect mirrors and clean or replace as necessary. • Call maintenance to correct the mode problem.
Bottom of the part is not cut through.	• Power is too low for feedrate.	• Increase power. • Decrease feedrate.

WOOD

PROBLEM	POSSIBLE CAUSES	REMEDY
Pierce Problems		
Loud "popping" sound during the pierce.	• Improper assist gas. • Power too high.	• Use nitrogen assist gas. • Reduce power.
Pierce not complete.	• Power too low. • Time too low.	• Increase pierce power. • Increase pierce time.
Edge quality problems (independent of location along part edge)		
Lower section of edge rough or uncut.	• Power too low. • Feedrate too high. • Focus too high.	• Increase cut power. • Reduce feedrate (approx 10%). • Lower the focus.
Edge quality problems (depends on location along part edge or size of cutout)		
Excessive burning in corners.	• Power too high at corner. • Feedrate in corners too low.	• Decrease dynamic power minimum. • Increase feedrate.
Bottom of part not cut through.	• Power is too low for feedrate.	• Increase power. • Decrease feedrate.
Deformed edge at transition from lead-in to contour.	• Power too high at transition. • Lead-in not optimal.	• Decrease dynamic power minimum. • Reduce cut power. • Adjust lead-in angle. • Adjust lead-in length. • Adjust lead-in feedrate.

LASER CUTTING DIAGNOSTIC WORKSHEET

Customer Information

Your name..... _____ Your phone number ..... _____

Date worksheet completed..... _____

Machine Information

Laser System Model ..... _____ Laser System S/N..... _____

Date of last PM service..... _____ Hours since last PM sevice ..... _____

Programming System..... _____ Resonator Size..... _____

Description of Cutting Problem

Material Type..... _____ Material Thickness..... _____

Describe the cutting problem:

Were any changes made to the Laser System or the cutting process about the same time this problem began?
Describe them:

Diagnostic Checklist

Is the correct nozzle being used (per the recommended parameters)?

Is the lens centered?

Is focus set properly (per the recommended parameters)?

Is the standoff set properly (per the recommended parameters)?

Is the correct feed rate being commanded?

Is the commanded power being delivered?

Is the commanded assist gas pressure being delivered?

Have you tried to solve the problem using the diagnostic charts provided?

If you have followed the diagnostic procedure and still cannot resolve the cutting problem, call:

CINCINNATI INCORPORATED – Laser Applications
(513) 202-5282

If poor cutting persists after trying the suggestions in the Process Notes section and the Diagnostic Guide section, the problem may be related to the three main laser beam characteristics – Power Stability, Beam Alignment, and Mode Quality. The procedures in this section explain how to measure and evaluate these characteristics. The information from these three procedures will help CINCINNATI Laser Service diagnose the problem efficiently and accurately.

CHECKING POWER STABILITY

Power stability can be easily checked, and is a good indicator of a properly operating laser system. To check the power stability, perform the following steps:

1. Select the menu items MAINT DIAGNOSTICS-RESONATOR-MANUAL POWER CONTROL.

2. Edit the screen as illustrated in Figure E-1 and press UPDATE.

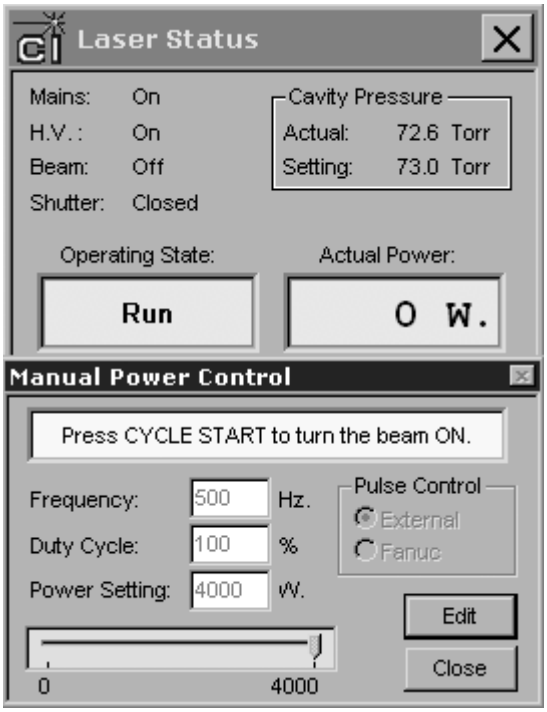


Figure E-1. Setup for Power Stability check.

3. Press CYCLE START.

4. Watch the displayed power in the remote terminal for 15 minutes. If the laser is operating normally, the displayed power will stay within 2% (80W) of 4000W. If the power fluctuates more than +/- 2%, contact CINCINNATI INCORPORATED Laser Service at (513) 367-7466 for further instructions.

5. Use the following table to record the power at the designated time intervals throughout the 15-minute test. Also, record the laser power calibration coefficient (Parameter #15204). Refer to the *Operation, Safety, and Maintenance Manual* for details on viewing machine parameters.

Beam ON Time (min)	Displayed Power (W)
0.5	
1	
2	
5	
10	
15	

CHECKING BEAM ALIGNMENT

FOUR-CORNER CARD SHOTS

! ! DANGER ! !

THIS PROCEDURE REQUIRES POTENTIALLY DANGEROUS ACTIONS, AND SHOULD NOT BE PERFORMED WITHOUT PROPER TRAINING.

1. Jog the upper pallet out, and the lower pallet in. Jog X and Y axes to the Xfar, Yfar position and the Z-axis to the full up position.

2. Turn the HV INTERLOCK switch to BYPASS.

3. Remove the cutting head.

4. Insert the cross hair assembly into the Z-axis aperture. Rotate the fixture so the card slot is parallel with the Y-axis and facing toward the load frame.

5. Select the menu items MAINT-MIRROR ALIGNMENT and set the mirror alignment screen to the following settings; length of flash – 0.020 sec, power level – 4000 W, duty cycle – 2%, frequency – 50 Hz.

6. Fold a thermal sensitive card (C.I. #923025) in half the long way, and tear along the fold to produce two cards approximately 1.6” (40.6 mm) wide by 6.5” (165.1 mm) long.

7. Place one card in the cross-hair fixture with the shiny side facing up. Position the card so the beam strikes the left side.

8. Jog the Z-axis to the up position.
9. Flash the shutter once.
10. Reposition the card for another shot.
11. Jog the Z-axis to the down position.
12. Flash the shutter once.
13. Repeat steps 8 through 12 to take the card shots at the other three table locations (2, 3, and 4), following the numerical order shown in Figure E-2.

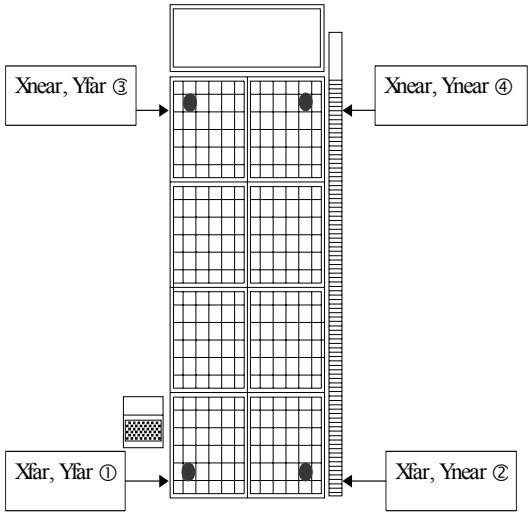


Figure E-2. Card shot positions in cutting area.

14. Label the cards as shown in Figure E-3.

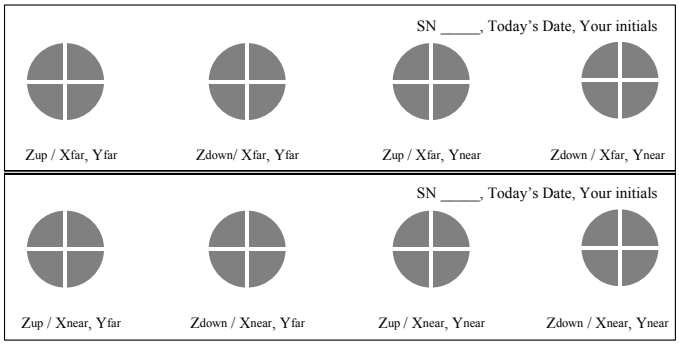


Figure E-3. Labeling of cards for card shots.

15. If all card shots are centered in the cross-hair image within +/- 0.060" (1.5 mm), the alignment is good. If the misalignment is over 0.060" (1.5 mm) in any of the shots, a beam alignment is required. Refer to the *Operation, Safety, and Maintenance Manual* for details about the beam alignment procedure.

CHECKING MODE QUALITY

! ! DANGER ! !

THIS PROCEDURE REQUIRES POTENTIALLY DANGEROUS ACTIONS, AND SHOULD NOT BE PERFORMED WITHOUT PROPER TRAINING.

1. Jog the upper pallet out, and the lower pallet in. Jog the X and Y axes to the Xfar, Yfar position and the Z-axis to the full up position.
2. Turn the HV INTERLOCK switch to BYPASS.
3. Remove the cutting head.
4. Lay a flat piece of mild steel under the Z-axis aperture.

CAUTION

Do not use Aluminum, Brass, or Copper. These materials are extremely reflective to the laser's wavelength.

5. Select the menu items MAINT-MIRROR ALIGNMENT and set the mirror alignment screen to the following settings; length of flash – 2.0 sec, power level – 4000W, duty cycle – 100%, frequency – 2000 Hz. Place a check next to the BEAM ON option. Set the warm-up settings as follows: power level – 4000W, duty cycle – 100%, frequency – 2000 Hz.
6. Place a mode block (C.I. #910623) under the Z-axis aperture. Align the block so the long side is parallel to the Y-axis, and position it so the burn will be placed on the leftmost portion of the mode block.
7. Position a hair dryer so the air is blowing across the block at about a 45-degree angle to the surface of the block.
8. Flash the shutter once for the full 2.0-second duration. The hair dryer will prevent flames from forming. If flames do form, release the shutter flash button and try again with a new mode block. It may be necessary to hold the hair dryer output closer to the burn to prevent flare-up. The resulting impression in the acrylic block is called a "Cold" mode burn because the resonator was not pre-heated prior to taking the burn.
9. Press CYCLE START to begin laser warm-up and move the block towards Y- for the second burn.

10. Allow approximately 3 minutes for the laser to stabilize.

11. Repeat the shutter flash as in Step 8.

12. Press CYCLE STOP to turn the beam off, and wait approximately 3 minutes to allow the laser to stabilize.

13. Jog the axes to the (Xnear, Ynear) position and repeat steps 6 through 12, except continue moving the block towards Y- for the remainder of the burns.

14. Examine the mode burns. A properly operating system will produce modes similar to those in Figure E-4. Refer to the next section for information about evaluating the table mode burns.

TABLE MODE BURN EXAMPLES

The table mode burns shown in Figures E-4 through E-10 were taken with the cutting head removed. Mode burns are evaluated on the basis of circularity, side view symmetry, size, and the presence of miscellaneous abnormalities.

Circularity

Circularity is judged by measuring the aspect ratio at a scribed depth of 0.120" (3.0 mm). A depth micrometer is a useful tool to make this scribe line. The aspect ratio is the ratio of the largest to smallest diameter measured at this depth. An aspect ratio greater than 1.2 is not acceptable. See Figures E-4 and E-5 for examples of good and bad circularity.

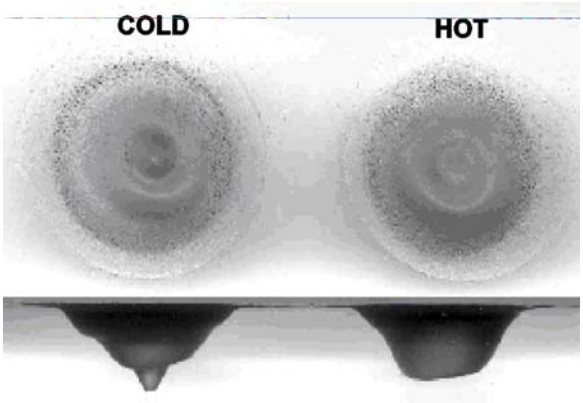


Figure E-4. Good side view symmetry and circularity.

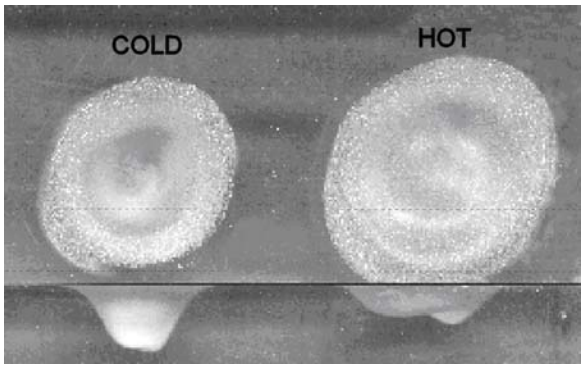


Figure E-5. Poor side view symmetry and circularity.

Side View Symmetry

Side view symmetry is not measured numerically. Ideally, the slope of the sidewalls of the mode burn is the same on all sides (see Figure E-4). Poor side view symmetry can be caused by resonator problems (see Figure E-5, hot mode) or by contamination of the beam delivery system (see Figure E-6).

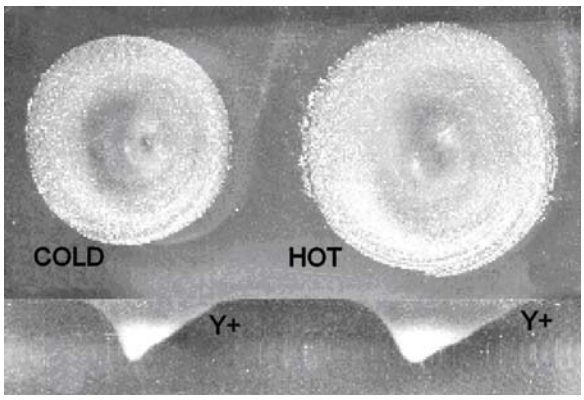


Figure E-6. Beam delivery contamination.

Resonator problems that produce poor side view symmetry include poor optical alignment, dirty optics, or damaged optics.

Beam delivery contamination can result from openings in the bellows, improper airflow from the beam purge, or a coolant leak in the beam delivery. Notice that the bottom of the mode in Figure E-6 has a distinct slope. When the beam delivery system is contaminated, the Y+ side of the mode will be shallow. Address this problem by checking all of the coolant fittings and hoses for leaks. Also, make sure the beam delivery is completely sealed and that there is adequate flow of clean dry air (or nitrogen) from the beam purge.

Mode Size

The mode size is measured by the full outer burn diameter of the mode, not the diameter at 0.120" (3.0 mm) depth. The typical size range for a hot table mode is 1.3 to 1.5" (34 to 38 mm). It is important to note that machines with mode diameters slightly out of this range

(that have good circularity and symmetry) may also perform well.

If the size of the hot mode is different from that of the cold mode, the laser internal optics might need to be cleaned or replaced. Some size difference is acceptable. Figure E-7 shows an extreme case of unacceptable mode diameter differences.

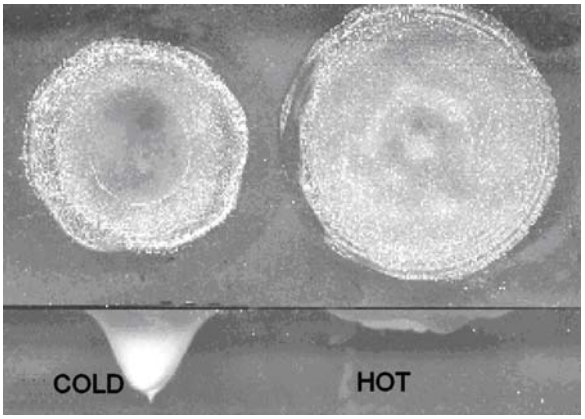


Figure E-7. Mode burn resulting from damaged internal optic.

Miscellaneous Abnormalities

Miscellaneous abnormalities include interference rings, double peaks, and stray energy on the periphery. These characteristics are shown in Figures E-8 through E-10.

Interference rings (see Figure E-8) are caused by "clipping" of the beam at an aperture in the beam delivery system. To find where the beam is clipping, the beam alignment should be checked at every external beam bending mirror.

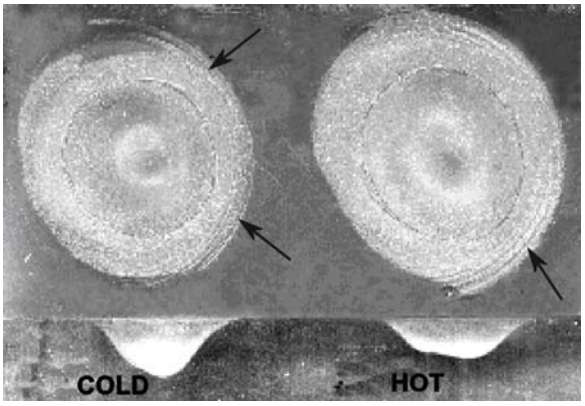


Figure E-8. Interference rings (right side of mode).

A double peak (see Figure E-9) is usually caused by misaligned, dirty, or damaged internal optics. To fix this problem, resonator optical alignment is required. During this procedure, the internal optics must be inspected and either be cleaned or replaced.

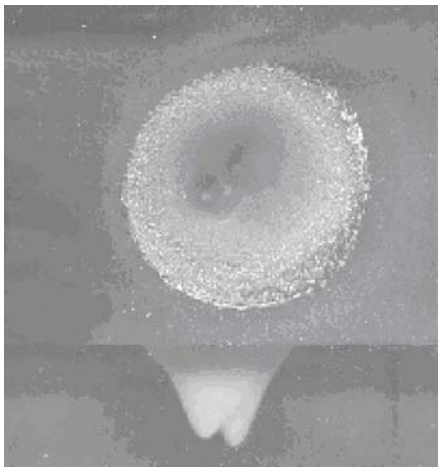


Figure E-9. Double peaks in mode.

A mode with stray energy around the periphery (see Figure E-10) is usually caused by misaligned, dirty, or damaged internal optics. To fix this problem, resonator optical alignment is required. During this procedure, the internal optics must be inspected and either be cleaned or replaced.

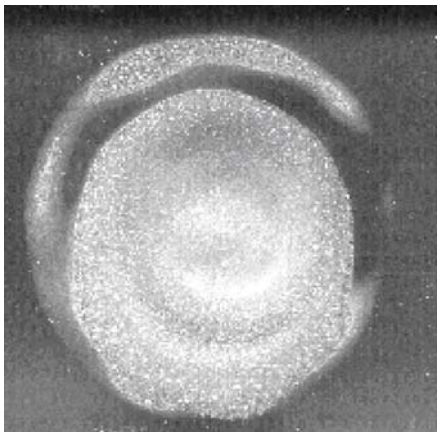


Figure E-10. Extra ring of energy in mode.

COPYRIGHT © 2005
CINCINNATI INCORPORATED